Fuzzy Logic and Control 2

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Last time

- Fuzzy set
- Membership function μ
  - Gives the degree of membership/certainty/truth
- Universe of discourse
  - The range of all possible values for an input to a fuzzy system.
- Rule-base
  - Linguistic rules/variables/values, antecedents (premises) and consequents (actions)
- Fuzzification, inference, defuzzification

Exercises

(vi) Draw a membership function that quantifies the statement “the number x is near 10.”

(vii) Draw a membership function that quantifies the statement “the number x is less than 10.”

(viii) Draw a membership function that quantifies the statement “the number x is greater than 10.”

Property size?

(iv)	Draw a membership function that quantifies the set of all small properties.
(v)	Draw a membership function that quantifies the set of all big properties.

Fuzziness vs. probability - which bottle would you drink?

A = { “Set of all potable (drinkable) liquids” }

a) Fuzzy bottle



$\mu_A(a) = 0.9$

b) Probabilistic bottle



$p(b \in A) = 0.9$

Motivations for fuzzy control

- the process to be controlled exhibits a **pronounced nonlinear behavior,**
- **no mathematical model** of the process is available because the modeling effort is unacceptably high or the process is not well understood,
- **expert knowledge** plays a key role in controlling the process and should be acquired and used for automatic control, or
- a multidimensional nonlinear relationship (e.g. a control law) should be represented such that **it can be understood and modified easily**.

Source: *Jaekel et al. 2014*

Is fuzzy logic and control useful?

- **Industry**: Heating, ventilation, and air conditioning (HVAC); washing machines; investment management; traffic control; …
- **Used in many current AI techniques**: Fuzzy clustering, fuzzy decision trees, fuzzy classification, regression and association rules, …
- **Science:** in 2023, Lotfi A. Zadeh was in the top 3 of most cited AI researchers (in 2024, he is 7th)

Notation for fuzzy sets

- A fuzzy set *A* is usually expressed in terms of its membership function μ_A which maps domain elements (x) their respective degrees of belonging in the interval [0,1]:

$$A = \{(x,\mu_A(x)) \mid x \in X\}$$

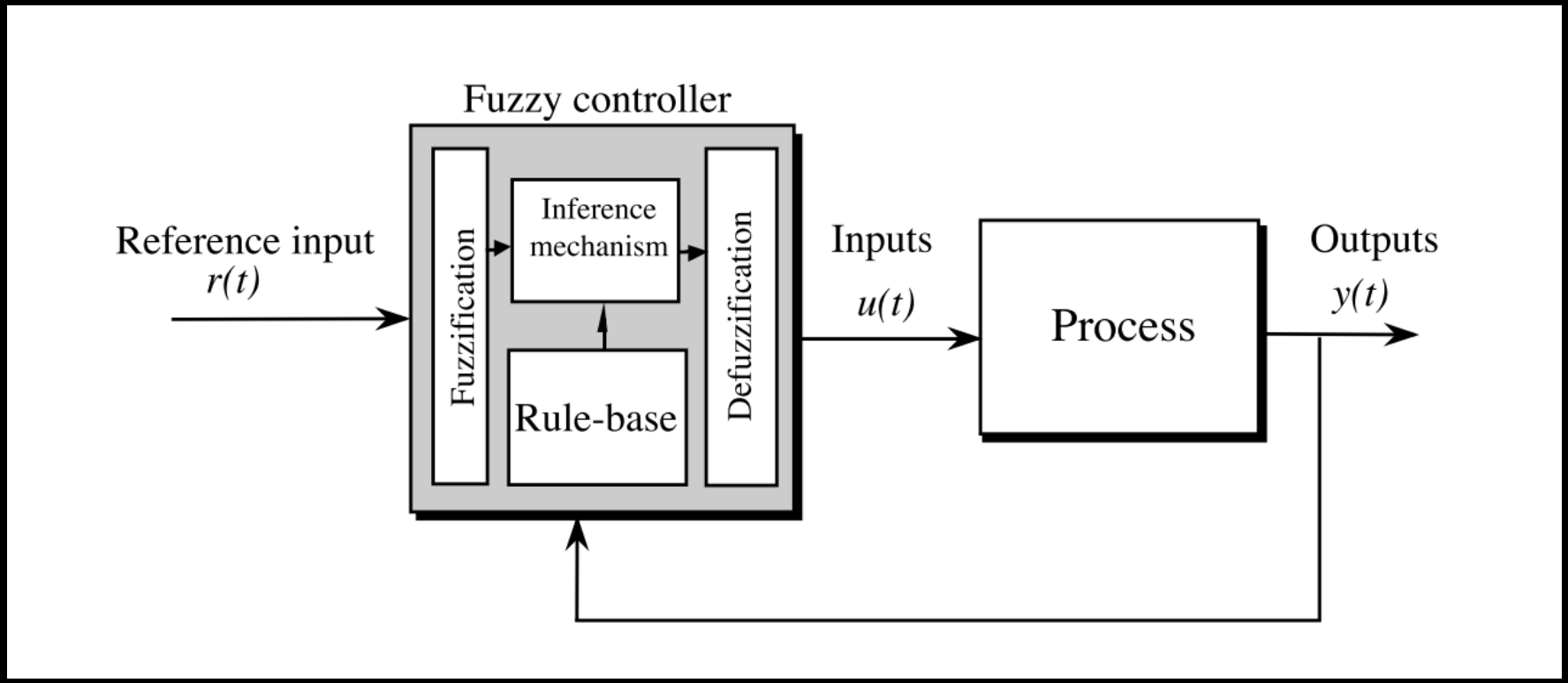
- Or using Zadeh's notation:

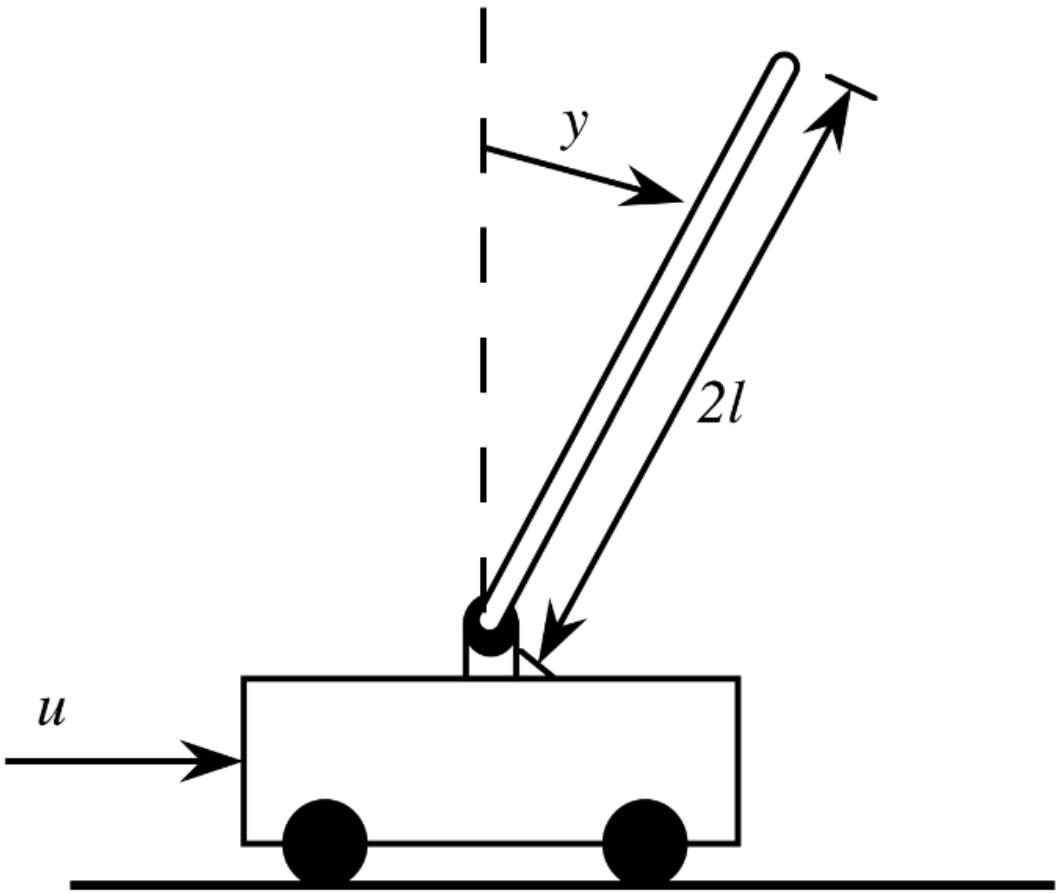
$$A = \{0.4/x_1 + 0.3/x_2 + 1/ x_3 + 0.6/ x_4\}$$

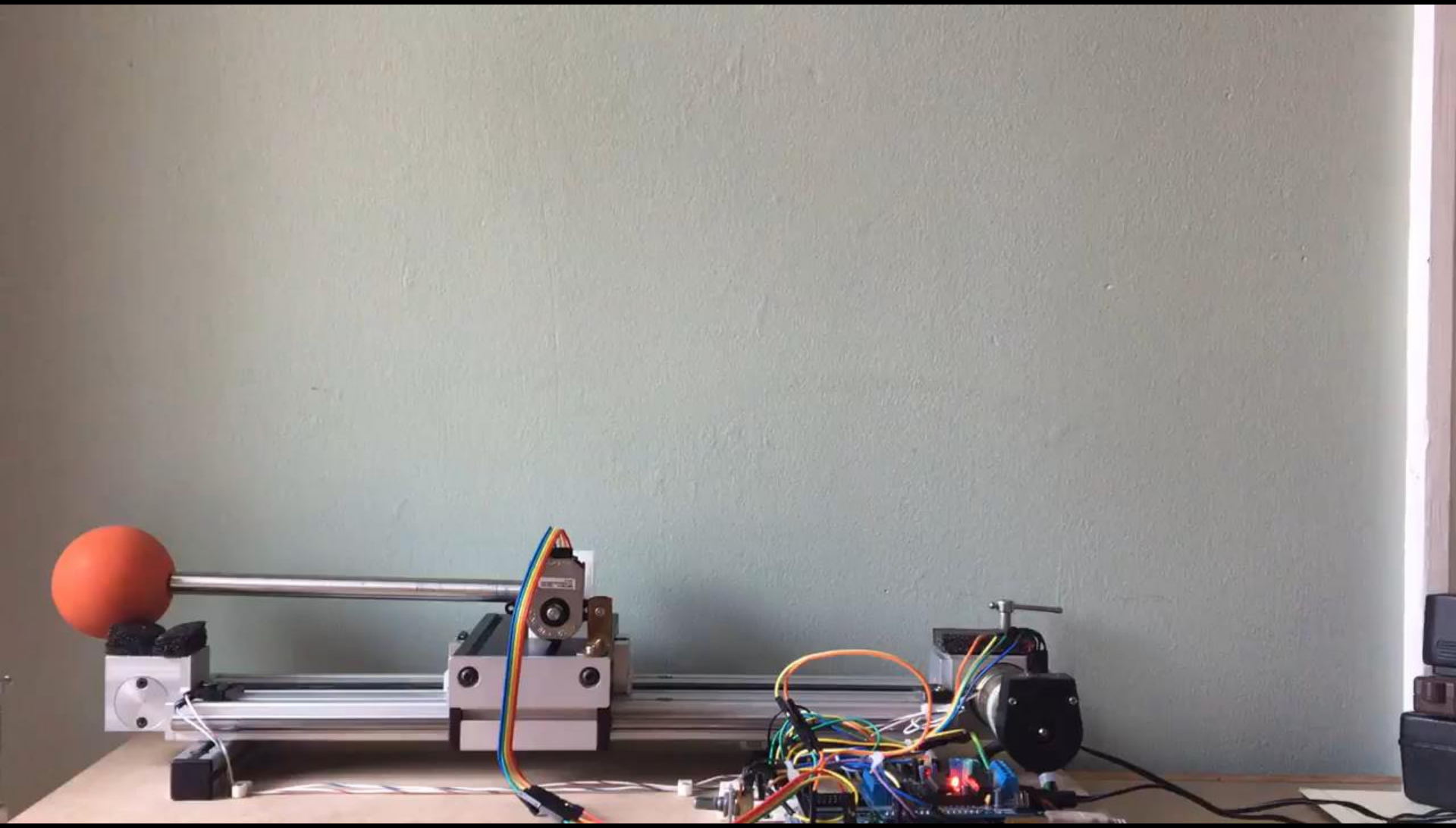
- A fuzzy set *A* over the universe for discourse *X* is a set of ordered pairs mapping domain elements to their respective degrees of belonging measured as a real number in the interval [0,1]:

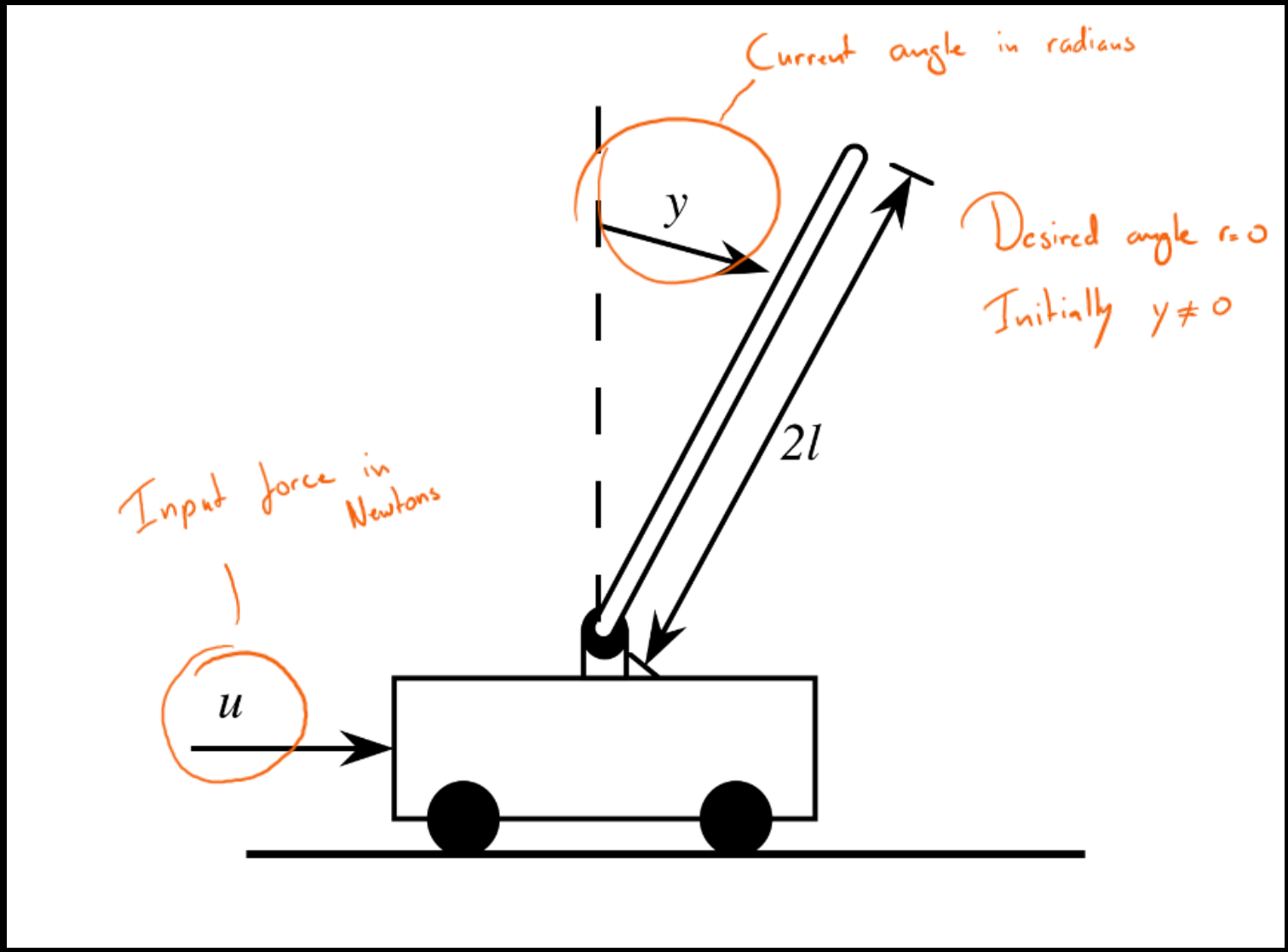
$$A = \{(x_1, 0.4), (x_2, 0.3), (x_3, 1), (x_4, 0.6)\}$$

Fuzzy control




y
2l
u



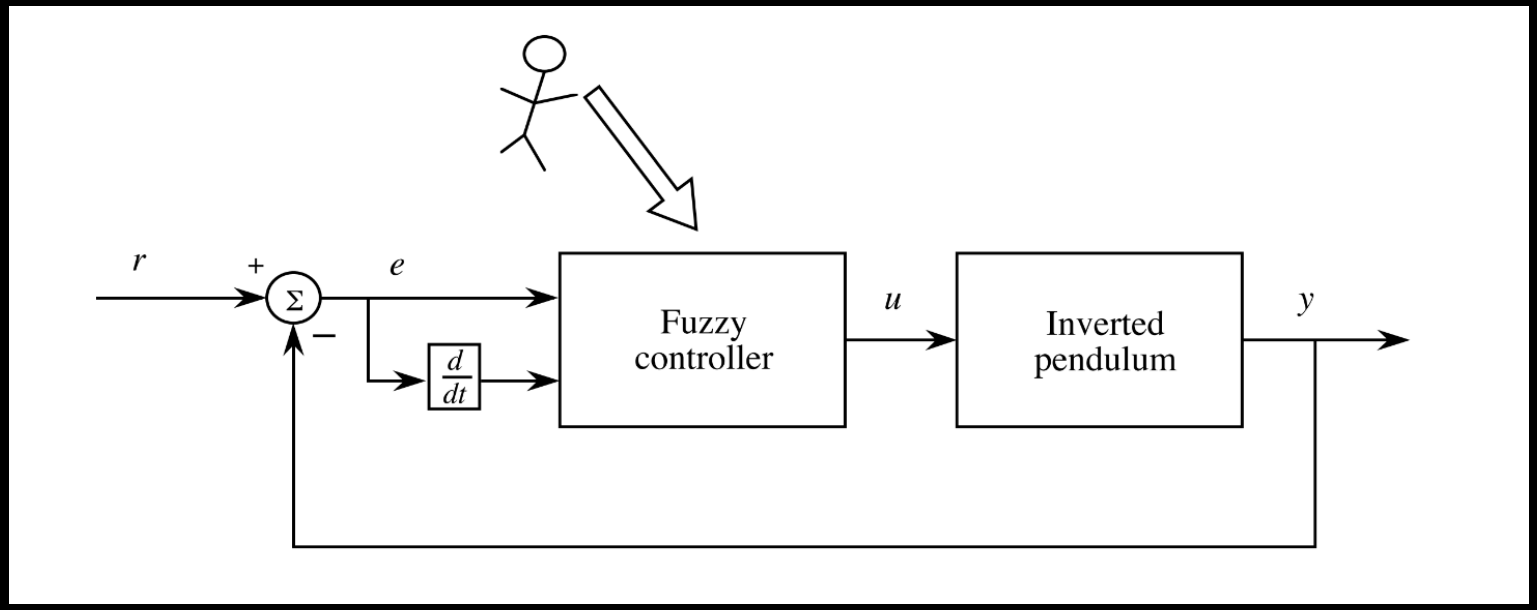

Current angle in radians
y
Desired angle r=0
Initially y≠0
2l
Input force in Newtons
u

Chosen inputs
(we could have
chosen others)

$$e(t) = r(t) - y(t)$$

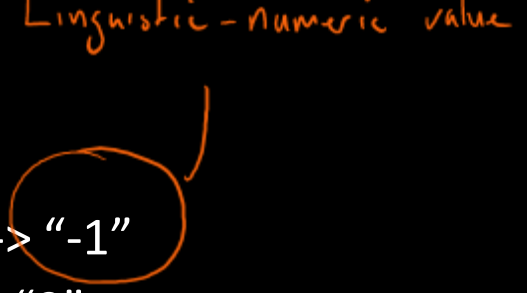
$$\frac{d}{dt}e(t)$$

We ask a human expert how
he/she controls the cart to
balance the pole



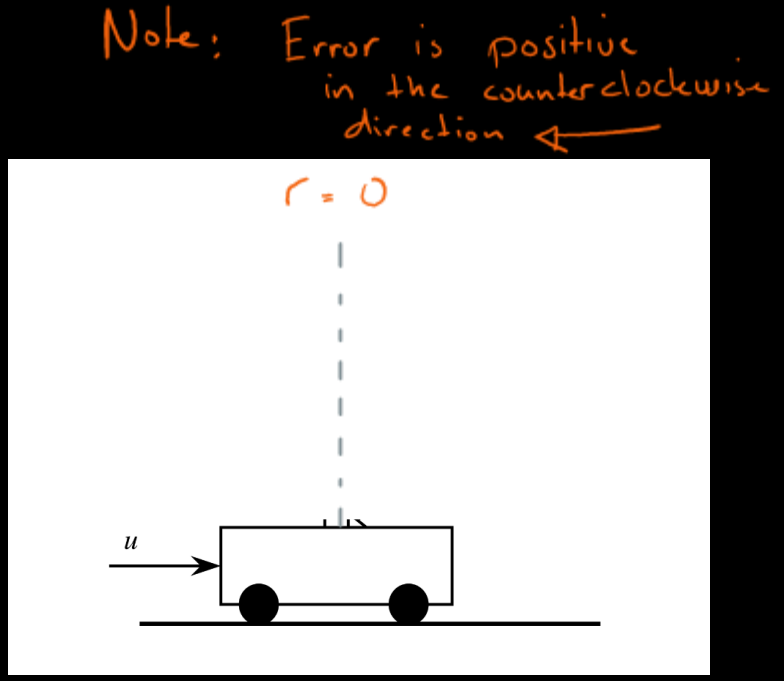
Linguistic descriptions

- Linguistic variables
  - "error" denotes $\mathrm{e}(t)$
    - $\mathrm{e}(t) = -y(t)$
  - "change-in-error" denotes $\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{e}(t)$
    - $\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{e}(t) = -\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{y}(t)$
  - "force" denotes $u(t)$



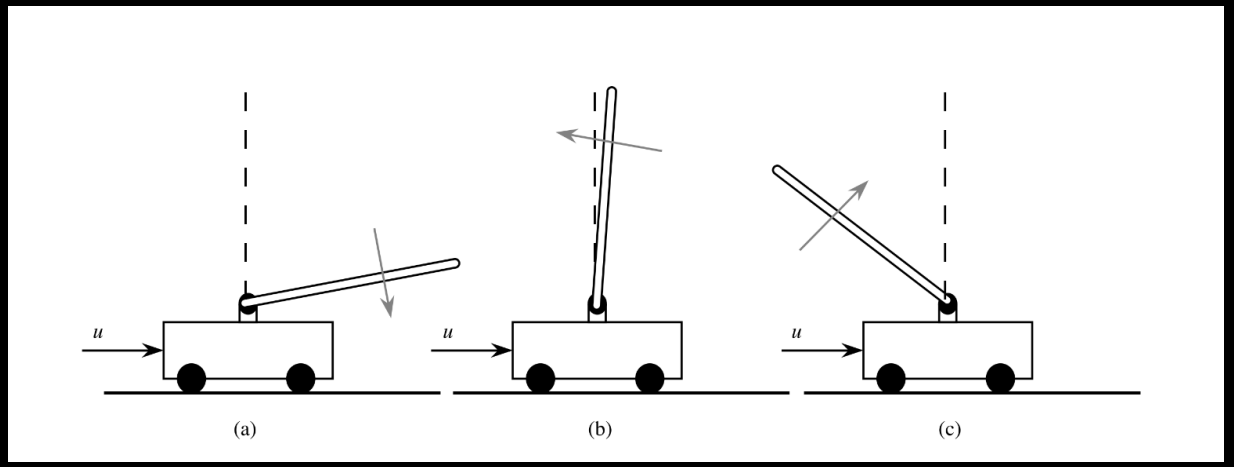
- Linguistic values
  - "negative small in size" --> "negsmall" --> "-1"
  - "positive large in size" --> "poslarge" --> "2"
  - Values used: "neglarge, negsmall, zero, possmall, poslarge" and "-2, -1, 0, 1, 2"

Example statements



- “error is poslarge”
- “error is negsmall”
- “error is zero”
- “error is neglarge and change-in-error is possmall”

Fuzzy rules



a) **If** error is neglarge **and** change-in-error is neglarge

Then force is poslarge

b) **If** error is zero **and** change-in-error is possmall

Then force is negsmall

c) **If** error is posslarge **and** change-in-error is negsmall

Then force is negsmall

Rule table

"force" u		"change-in-error" $\dot{e}$				
		-2	-1	0	1	2
"error" e	-2	2	2	2	1	0
	-1	2	2	1	0	-1
	0	2	1	0	-1	-2
	1	1	0	-1	-2	-2
	2	0	-1	-2	-2	-2

The numbers in the table are linguistic-numerical values and the rule-base is therefore not precise, but represents abstract knowledge about how to control the cart.

Rule table

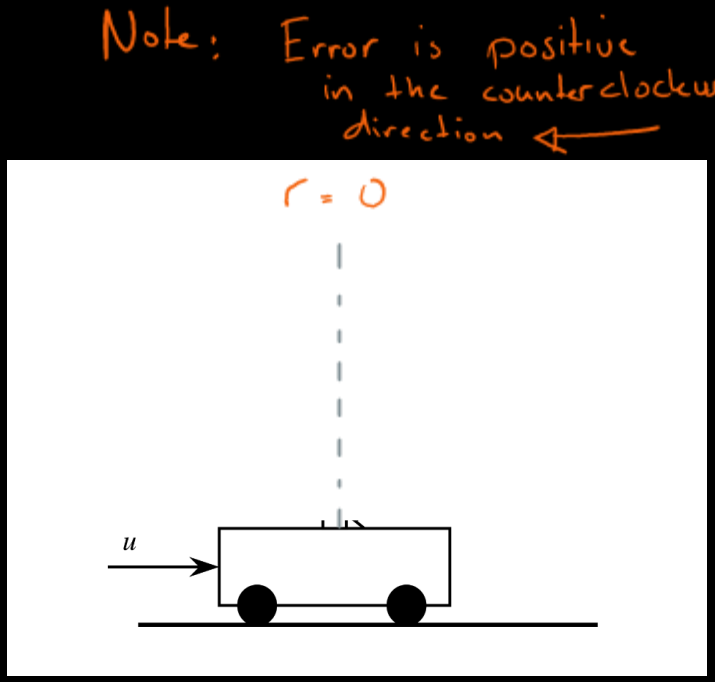
5 x 5 possible premises = 25 possible rules

"force" u		"change-in-error" $\dot{e}$				
		-2	-1	0	1	2
"error" e	-2	2	2	2	1	0
	-1	2	2	1	0	-1
	0	2	1	0	-1	-2
	1	1	0	-1	-2	-2
	2	0	-1	-2	-2	-2

The numbers in the table are linguistic-numerical values and the rule-base is therefore not precise, but represents abstract knowledge about how to control the cart.

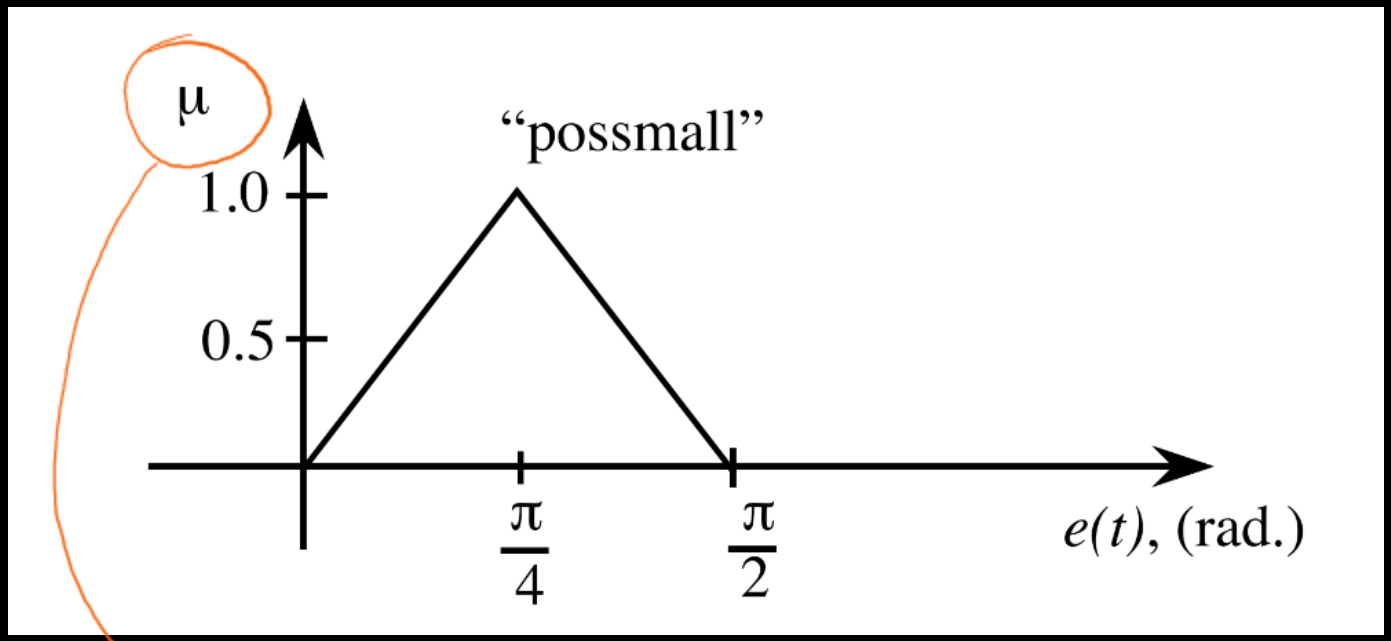
Quantification

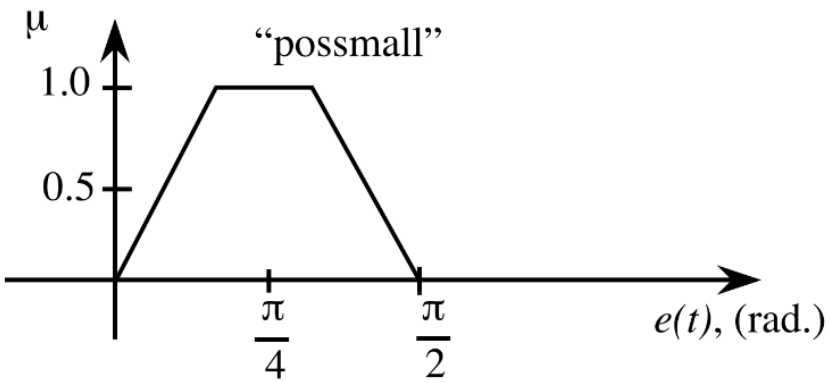
- When is “error” “possmall”?



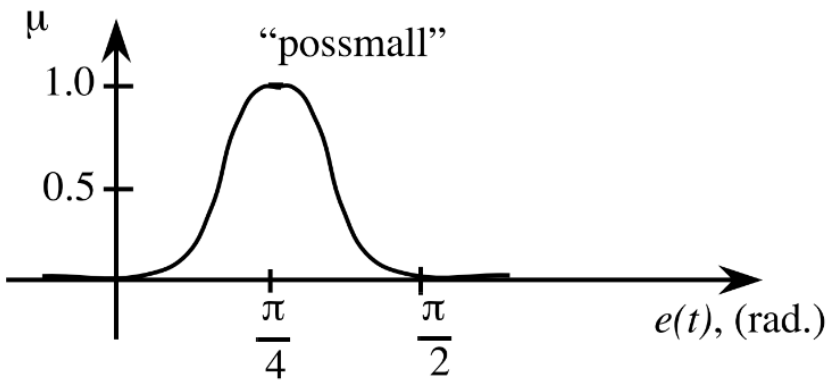
Quantification

- Membership function for “possmall” with respect to “error”:

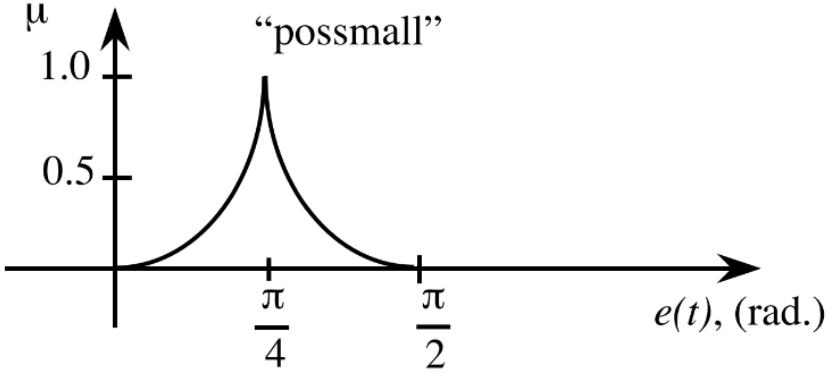




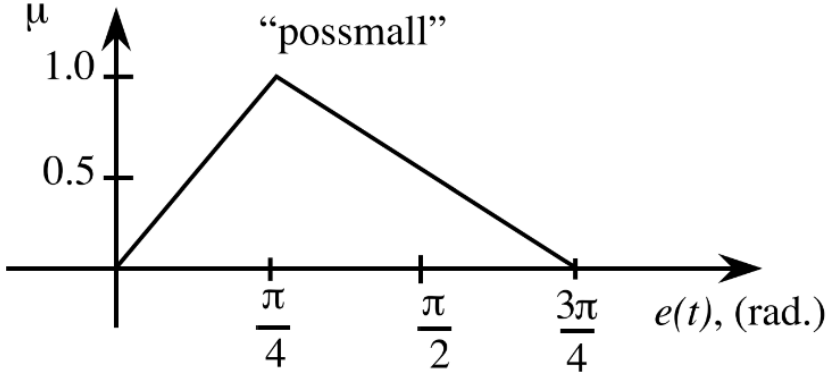
(a) Trapezoid.



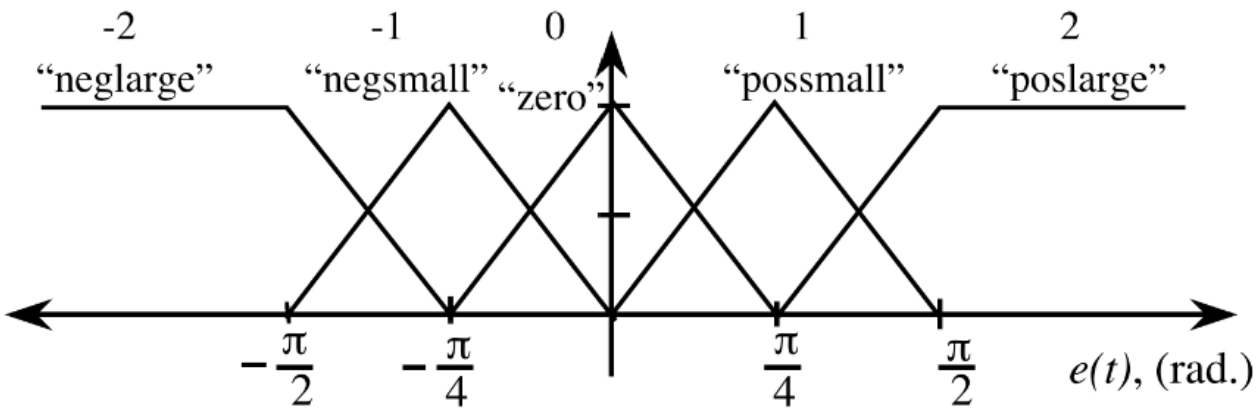
(b) Gaussian.

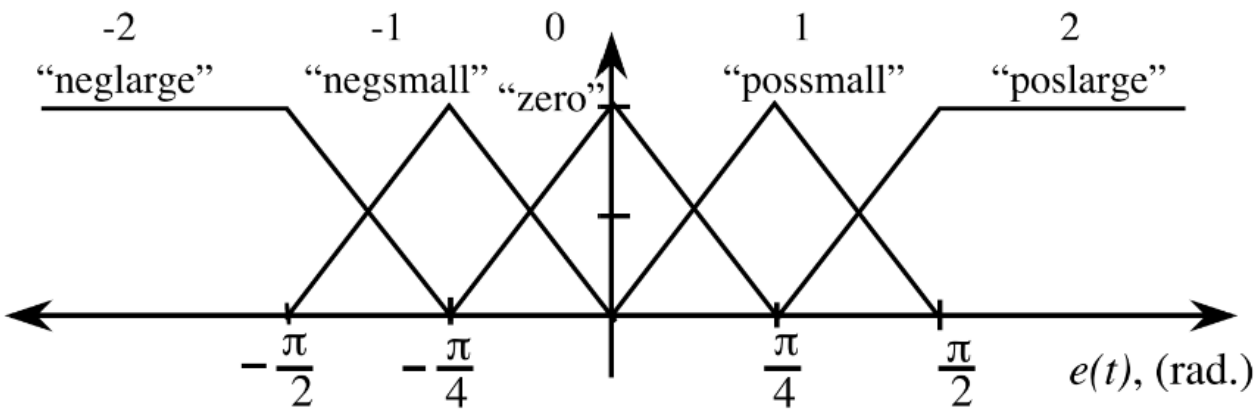


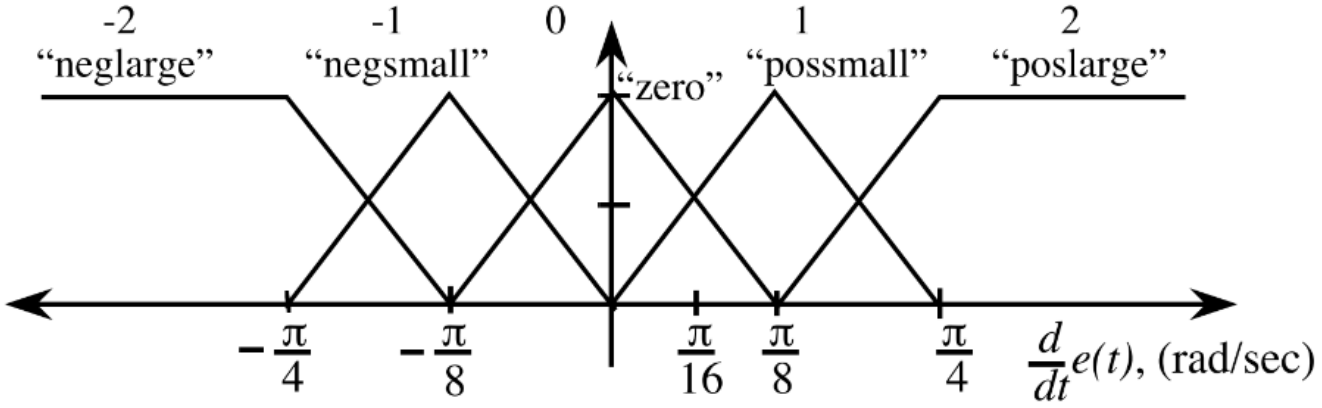
(c) Sharp peak.

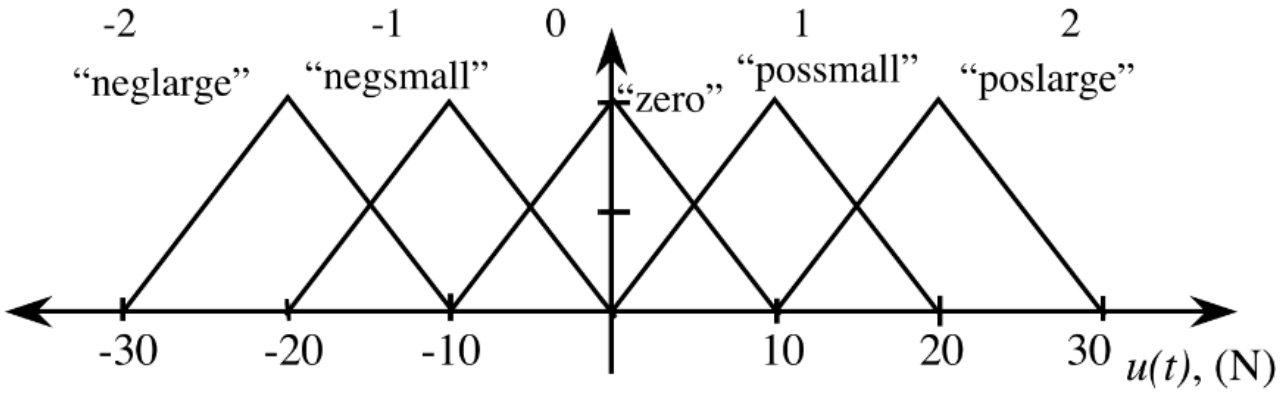


(d) Skewed triangle.

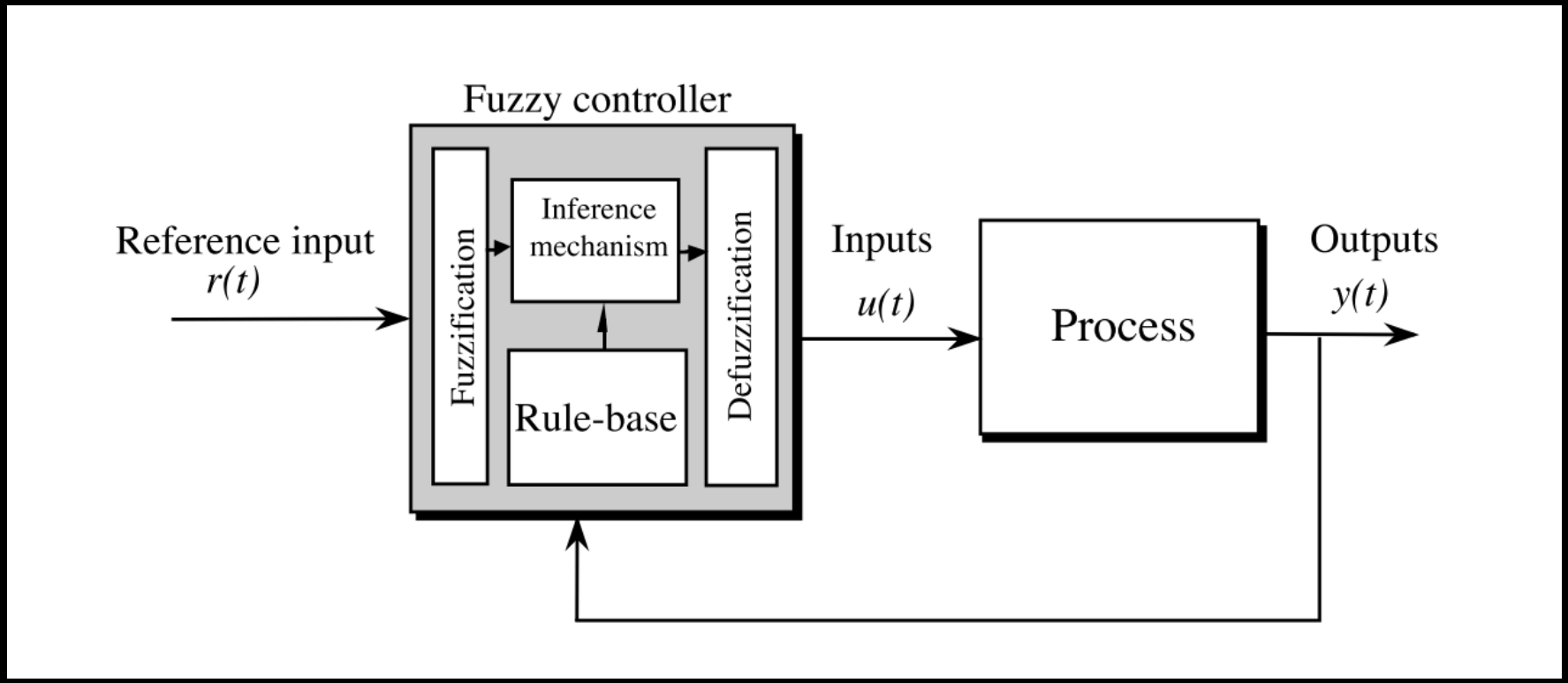

-2
-1
0
1
2
“neglarge”
“negsmall”
“zero”
“possmall”
“poslarge”
$-\frac{\pi}{2}$
$-\frac{\pi}{4}$
$\frac{\pi}{4}$
$\frac{\pi}{2}$
e(t), (rad.)


-2
-1
0
1
2
"neglarge"
"negsmall"
"zero"
"possmall"
"poslarge"
$-\frac{\pi}{2}$
$-\frac{\pi}{4}$
$\frac{\pi}{4}$
$\frac{\pi}{2}$
$e(t)$, (rad.)


-2
-1
0
1
2
"neglarge"
"negsmall"
"zero"
"possmall"
"poslarge"
$-\frac{\pi}{4}$
$-\frac{\pi}{8}$
$\frac{\pi}{16}$
$\frac{\pi}{8}$
$\frac{\pi}{4}$
$\frac{d}{dt}e(t)$, (rad/sec)


-2
-1
0
1
2
"neglarge"
"negsmall"
"zero"
"possmall"
"poslarge"
-30
-20
-10
10
20
30
$u(t)$, (N)

Fuzzy control

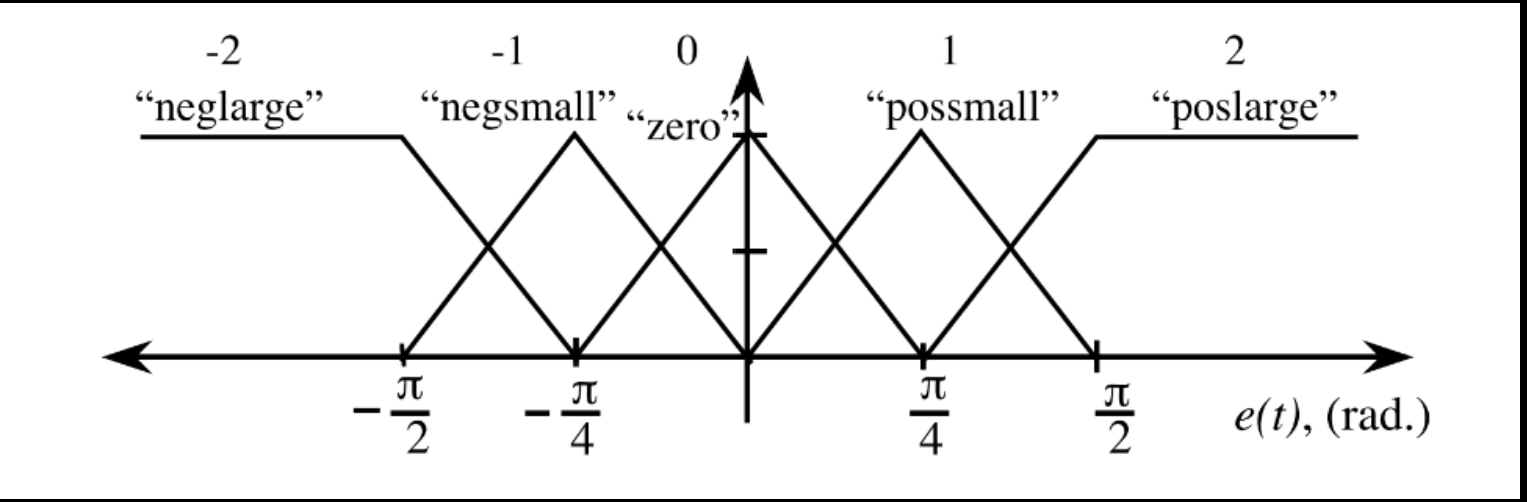


Fuzzification

- Map crisp input values to fuzzy sets

$$\tilde{S}(x) = \sum_{i \in S} \mu_i(x)/\mathrm{i}$$

- Example



$\tilde{S}(0) =$ { 0/neglarge + 0/negsmall + 1/zero + 0/possmall + 0/poslarge }

$\tilde{S}(\frac{\pi}{8}) =$ { 0/neglarge + 0/negsmall + 0.5/zero + 0.5/possmall + 0/poslarge }

Matching

Figuring out which rule(s) in our rule-base should be activated – and with what certainty.

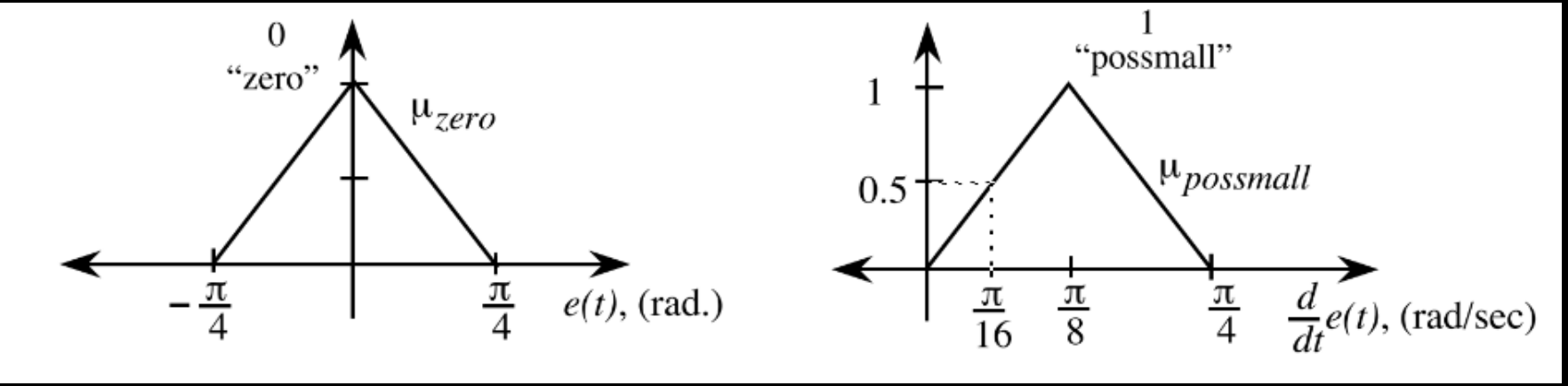
Example:

If error is zero **and** change-in-error is possmall **Then** force is negsmall

If the membership degree of error with respect to the "zero" fuzzy set is 0 ($\mu_{zero}(e(t)) = 0$) or the membership degree of change-in-error with respect to the negsmall fuzzy set is 0 ($\mu_{negsmall}\left(\frac{d}{dt}e(t)\right) = 0$), it is easy – the rule is not activated.

Matching

If error is zero **and** change-in-error is possmall **Then** force is negsmall



$$\mathrm{e}(t) = \frac{\pi}{8}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{e}(t) = \frac{\pi}{32}$$

$\mu_{zero}(\mathrm{e}(t))$ = _____

$\mu_{possmall}(\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{e}(t))$ = _____

Matching

$$\mu_{zero}(\mathrm{e}(t)) = 0.5 \qquad \mu_{possmall}\left(\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{e}(\mathrm{t})\right) = 0.25$$

If error is zero and change-in-error is possmall **Then** force is negsmall

What should we do about this "and"?

Matching

$$\mu_{zero}(\mathrm{e}(t)) = 0.5 \qquad \mu_{possmall}\left(\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{e}(t)\right) = 0.25$$

If error is zero and change-in-error is possmall **Then** force is negsmall

What should we do about this "and"?

Minimum: $\mu_{premise}\left(\mu_{zero}(\mathrm{e}(t)) \text{ and } \mu_{possmall}(\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{e}(t))\right) = \min(0.5, 0.25) = 0.25$

Product: $\mu_{premise}\left(\mu_{zero}(\mathrm{e}(t)) \text{ and } \mu_{possmall}(\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{e}(t))\right) = 0.5 \cdot 0.25 = 0.125$

T-norm:

The t-norm or "triangle norm" generalizes two-valued logic conjunction. Several others than the "Minimum t-norm" and the "Product t-norm" shown here exist.

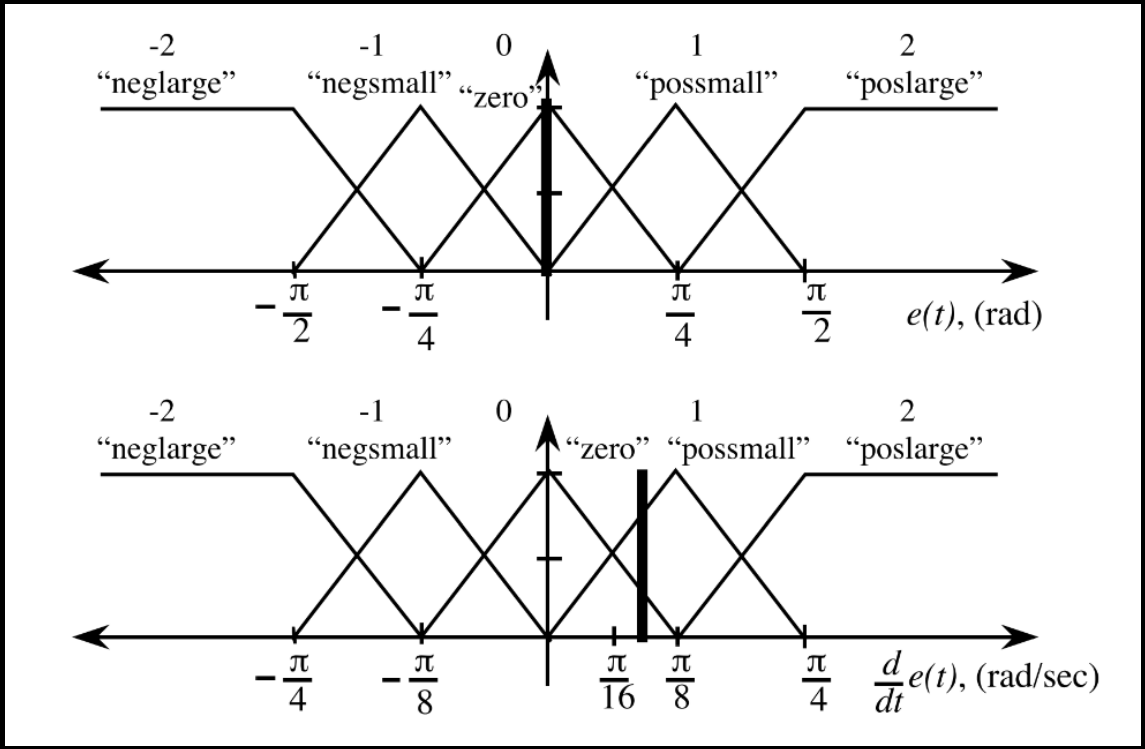
Matching

Crisp inputs: $e(t) = 0$ $\quad \frac{d}{dt}e(t) = \frac{3\pi}{32}$

$\mu_{zero}(e(t))$ = 1

$\mu_{zero}(\frac{d}{dt}e(t))$ = 0.25

$\mu_{possmall}(\frac{d}{dt}e(t))$ = 0.75



Hence, the following premise terms are "on":

"error is zero" (error is "0")

"change-in-error is zero" (change-in-error is "0")

"change-in-error is possmall" (change-in-error is "1")

Hence, the following premise terms are on:

"error is zero" (error is "0")

"change-in-error is zero" (change-in-error is "0")

"change-in-error is possmall" (change-in-error is "1")

"force" u		"change-in-error" $\dot{e}$				
		-2	-1	0	1	2
"error" e	-2	2	2	2	1	0
	-1	2	2	1	0	-1
	0	2	1	0	-1	-2
	1	1	0	-1	-2	-2
	2	0	-1	-2	-2	-2

Inference

Two rules are activated:

Rule 1:

If error is zero and change-in-error is zero **Then** force is zero

1

0.25

$\mu_{premise(1)} = \min(1, 0.25) = 0.25$

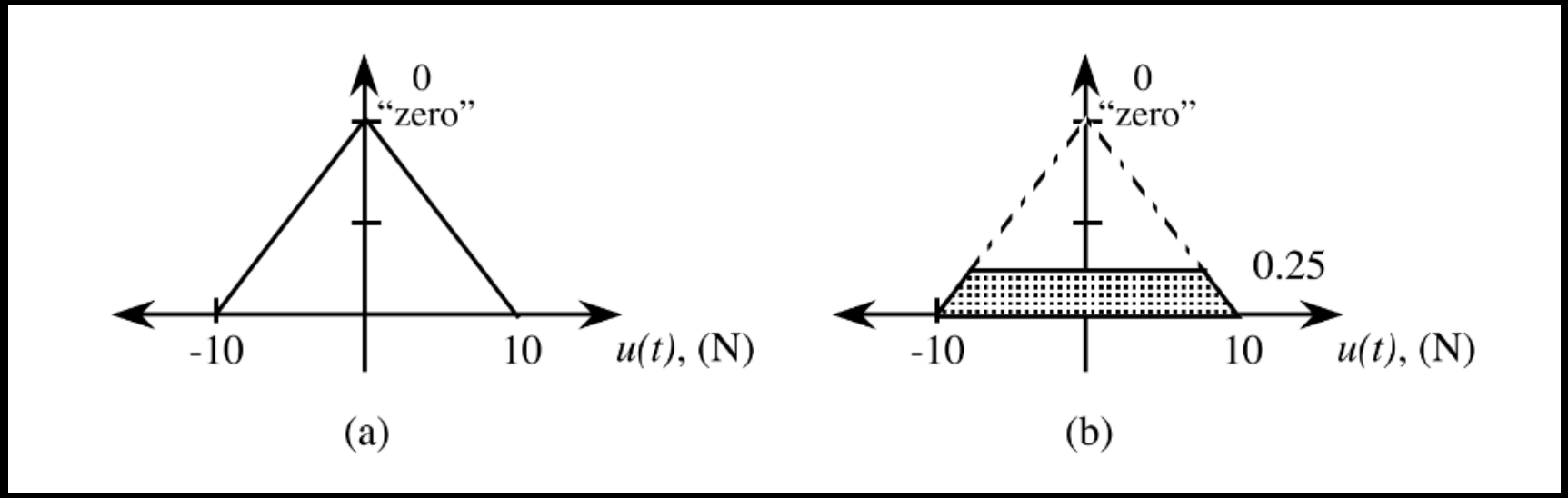
Recommendation of rule 1:

$\mu_{action(1)}(u) = \min(\mu_{premise(1)}, \mu_{zero}(u))$

Notice that we don't know u, the output, yet

$$\mu_{action(1)}(u) = \min(\mu_{premise(1)}, \mu_{zero}(u))$$

$\mu_{zero}(u)$ $\mu_{action(1)}(u)$



Inference

Two rules are activated:

Rule 2:

If error is zero and change-in-error is possmall **Then** force is negsmall

1 0.75

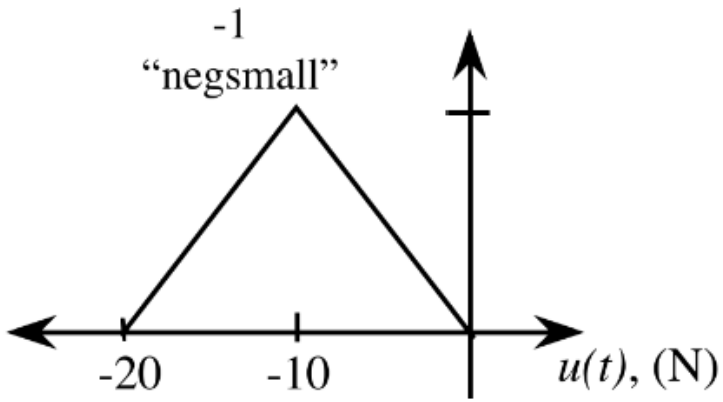
$\mu_{premise(2)} = \min(1, 0.75) = 0.75$

Recommendation of rule 2:

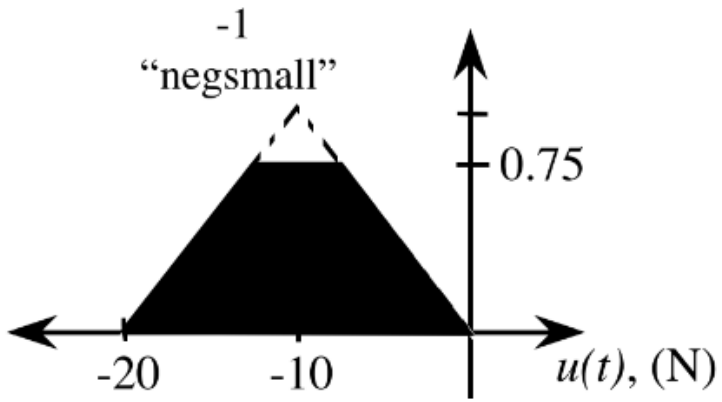
$\mu_{action(2)}(u) = \min(\mu_{premise(2)}, \mu_{negsmall}(u))$

$$\mu_{action(2)}(u) = \min(\mu_{premise(2)}, \mu_{negsmall}(u))$$

$\mu_{negsmall}(u)$

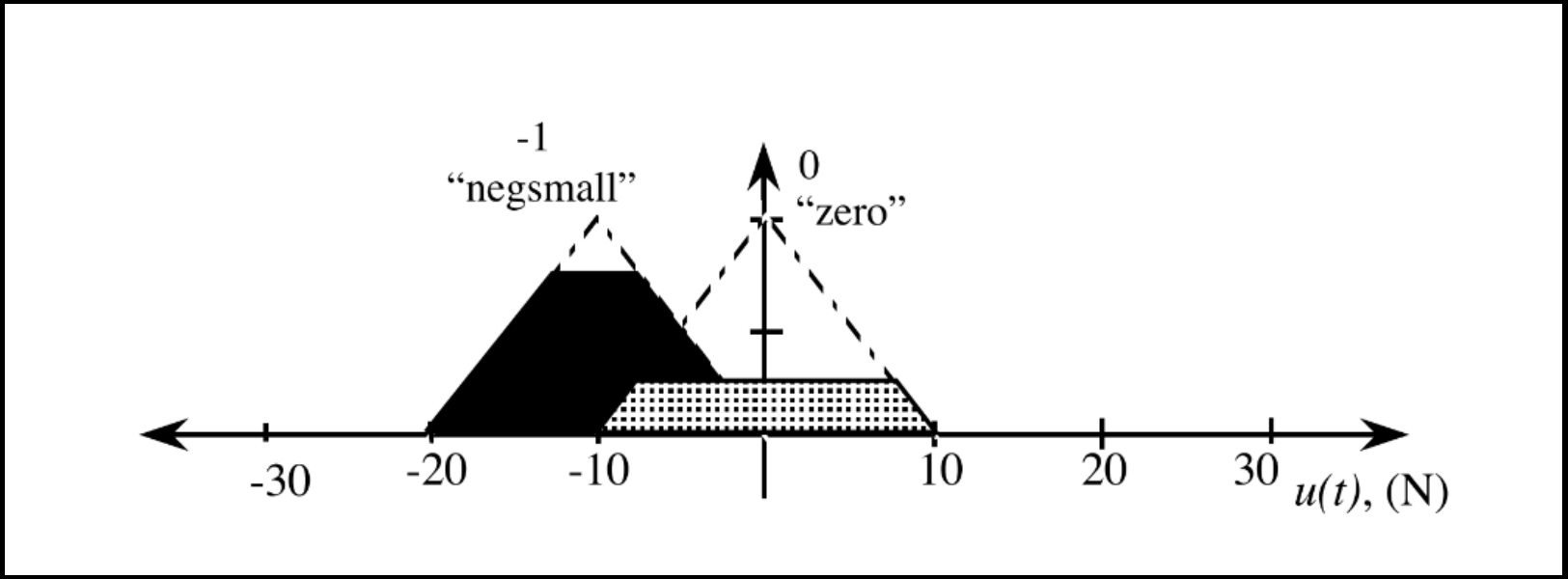


$\mu_{action(2)}(u)$

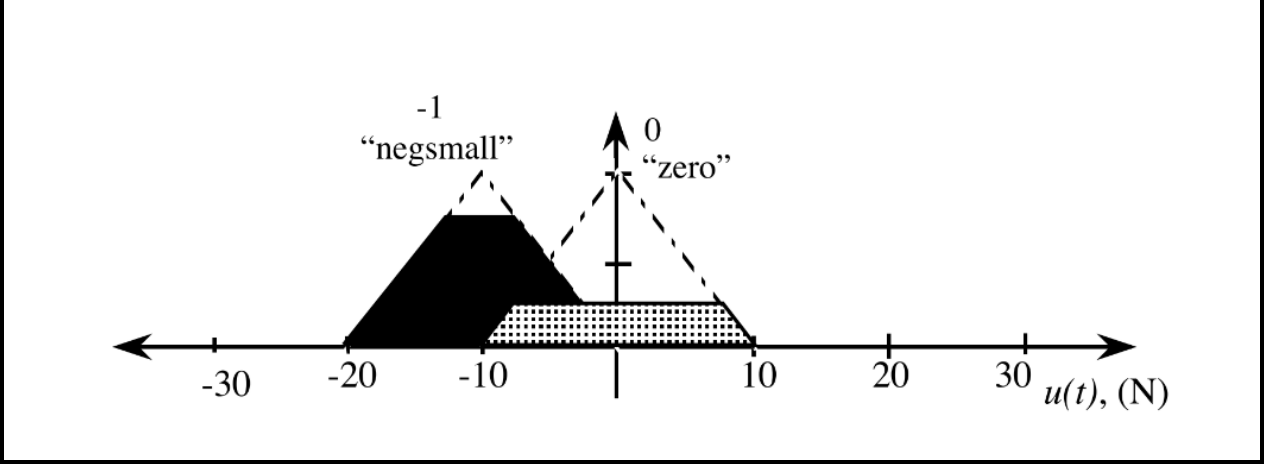


Defuzzification, turn fuzzy sets into crisp values

Rule 1 and 2 implied the following two fuzzy sets



Defuzzification



Center of Gravity (COG):

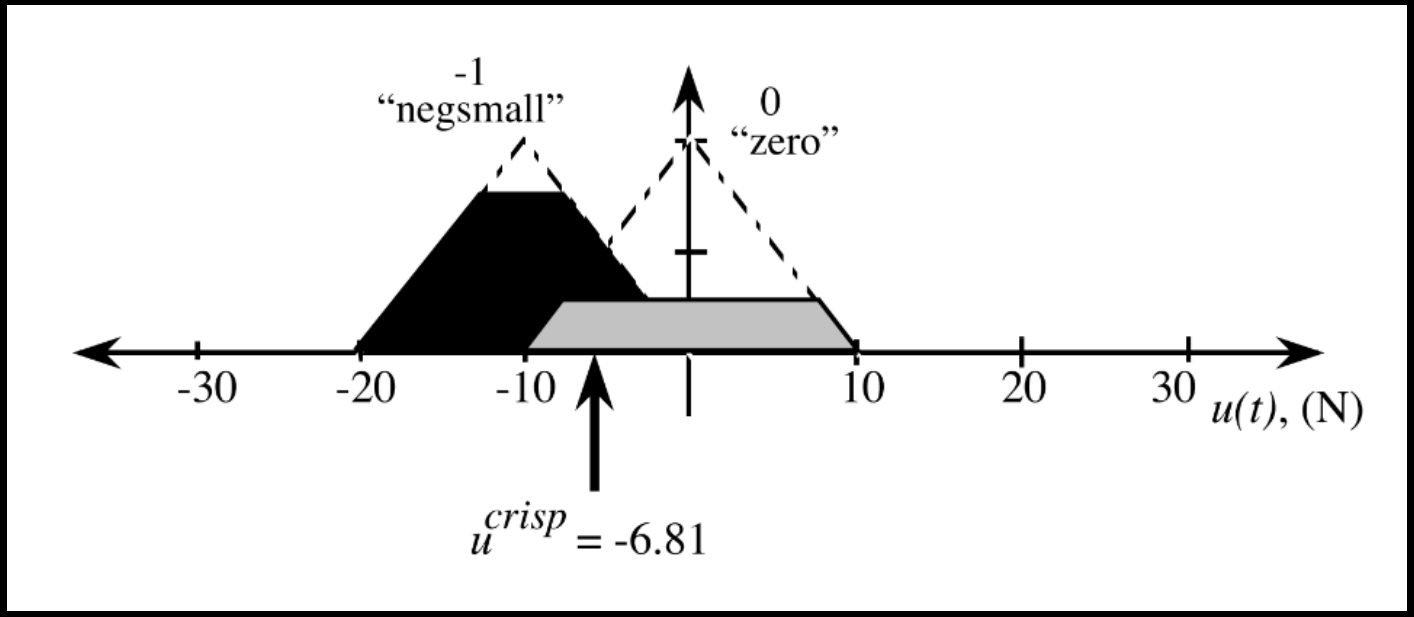
Let b_i denote the center of the membership function of the action of rule i. Hence, b_0 = 0 and $b_1 = -10$. Then the crisp output value is computed as follows:

$$u^{crisp} = \frac{\sum_i b_i \int \mu_{(i)}}{\sum_i \int \mu_{(i)}}$$

Defuzzification

$$u^{crisp} = \frac{\sum_i b_i \int \mu_{(i)}}{\sum_i \int \mu_{(i)}}$$

$$u^{crisp} = \frac{(0)(4.375) + (-10)(9.375)}{4.375 + 9.375} = -6.81$$



Summary

Crisp inputs:

$$e(t) = 0$$

$$\frac{d}{dt}e(t) = \frac{3\pi}{32}$$

